CUSTOMER SATISFACTION ANALYSIS

CSAT Analytics Project Documentation | Power BI

1. Project Overview

The purpose of this project is to understand customer support activity and review customer satisfaction using call records. The analysis focuses on call volume, contact channels, customer locations, reasons for calling, and response-time performance.

Dataset: project_csat_analytics CSAT
Rows loaded: 32,941
Columns: 13
Period: October 2020
Tool: Microsoft Power BI

The assignment description mentioned 32,941 rows and 12 columns. The connected dataset contained 32,941 rows and 13 columns. This document uses the values actually loaded into Power BI.

2. Data Preparation

- Connected Power BI to the MySQL database using the database details provided for the project.
- Loaded the project_csat_analytics CSAT table into Power BI.
- Reviewed column names and checked that dates, numbers, and text fields had suitable data types.
- Standardized the separators in call_timestamp and converted the field to Date using the English (United States) locale.
- Checked column quality in Power Query. All columns showed 100% valid values, with no errors or empty values.
- Checked for exact duplicate rows across all 13 columns. No exact duplicates were found; the row count remained 32,941.

3. Steps Followed to Create the Dashboard

1. **Connect to the database:** In Power BI Desktop, select the MySQL database connector, enter the host, port, database, and login details provided for the project, and connect to the CSAT table.

2. **Prepare the data:** Open Power Query, review fields and data types, correct the call date format, and check data quality and duplicate rows.

3. **Create KPI cards:** Add card visuals for Total Calls, Average CSAT, and Average Call Duration using the relevant fields.

4. **Create the call trend line chart:** Add a line chart. Put call_timestamp on the X-axis and call count on the Y-axis to show daily call volume for October 2020.

5. **Create the channel doughnut chart:** Add a doughnut chart. Use channel as the category and call count as the value.

6. **Create the state map:** Add a map visual. Use state as the location and call count to represent volume by state. Set the state data category appropriately.

7. **Create the reason column chart:** Add a column chart with reason as the category and call count as the value.

8. **Create the response-time bar chart:** Add a bar chart with response_time as the category and call count as the value.

9. **Add slicers:** Add slicers for Call Date, State, Sentiment, and Call Channel.

10. **Review and save:** Check titles, fields, labels, and slicer interactions. Save the Power BI report.

4. Dashboard Components

Visual	Purpose
KPI cards	Total Calls, Average CSAT, and Average Call Duration.
Line chart	Daily call volume trends across October 2020.
Doughnut chart	Share of calls received through each channel.
Map	Call volume by state.
Column chart	Number of calls by reason.
Bar chart	Number of calls by response-time category.
Slicers	Filters for Call Date, State, Sentiment, and Call Channel.

5. Key Findings

- **Call volume varied during October.** October 21 had the highest daily volume shown, with 1,170 calls. October 31 had the lowest, with 1 call.

- **Call-Center was the most-used channel.** It accounted for 32.30% of calls. Web was the least-used channel at 19.96%. Chatbot accounted for 25.06%, and Email accounted for 22.68%.

- **Billing questions were the main reason for calls.** The chart showed approximately 23.5K billing-related calls, compared with approximately 4.7K each for Payments and Service Outage. These are approximate chart readings.

- **Most calls were recorded as Within SLA.** This category contained 20,625 calls, or 62.61% of the total.

- **Some calls were recorded outside the Within SLA category.** There were 8,148 calls Below SLA (24.75%) and 4,168 Above SLA (12.65%). Interpret these labels according to the project's SLA definitions.
- **Three state values checked on the map were substantial.** California showed 3,631 calls, Texas 3,572, and Florida 2,834. These are verified examples, not a confirmed ranking of the top three states.

6. Three Important Recommendations

11. **Reduce billing-related customer issues:** Review common billing questions and identify confusing information. Make bills and payment instructions easier to understand, provide clear billing FAQs, and investigate repeated billing complaints. This may reduce avoidable repeat contacts.

12. **Improve response-time and SLA performance:** Track calls in the Below SLA and Above SLA categories. Review when delays occur, check whether staffing matches busy periods, and monitor response performance regularly. Focus improvement efforts on the situations where delays are most common.

13. **Improve the quality of customer support:** Review customer feedback and recurring complaints, provide focused agent training, and check whether chatbot and website support help customers resolve simple issues. Monitor CSAT results to see whether service changes improve the customer experience.

7. Conclusion

The dashboard brings together call trends, contact channels, state-level call volume, reasons for calling, and response-time categories. Billing questions stand out as the largest reason for customer calls. The response-time results also show that a meaningful share of calls were not recorded as Within SLA. The company can use these findings to review billing communication, monitor response performance, and improve customer support.